

PPA – Polyphthalamide PPA GF40

AKROMID® T5 GF 40 8 natural (6463)

Tensile modulus

15000 MPa

1 mm/min

ISO 527-2

Stress at break

245 MPa

5 mm/min

ISO 527-2

Charpy impact strength

80 kJ/m²

23°C

ISO 179-1/1eU

AKROMID® T5 GF 40 8 natural (6463) is a 40% glass fibre reinforced polyphthalamide with very high rigidity and strength, as well as high temperature and chemical resistance.

This aromatic PPA keeps mechanical performance even at elevated temperatures or moisture pick-up.

The material fulfills the requirements according to KTW, DVGW W270, ACS, WRAS as well as NSF. It is suitable for applications that are in contact with cold and hot drinking water as well as food.

Typical applications

Mainly components in domestic and industrial applications that are in contact with food or drinking water, requiring strength and durability even at high temperatures or moist conditions.

Regulatory**Mechanical Properties**

Tensile modulus (1 mm/min | ISO 527-2)

d.a.m.

15000 MPa

conditioned

15000 MPa

Stress at break (5 mm/min | ISO 527-2)

d.a.m.

245 MPa

conditioned

230 MPa

Strain at break (5 mm/min | ISO 527-2)

d.a.m.

2,4 %

conditioned

2,4 %

Charpy impact strength (23°C | ISO 179-1/1eU)

d.a.m.

80 kJ/m²

conditioned

80 kJ/m²

Charpy notched impact strength (23°C | ISO 179-1/1eA)

d.a.m.

11 kJ/m²

conditioned

11 kJ/m²**Thermal Properties**

Temperature of deflection under load HDT/C (8 MPa | ISO 75)

190 °C

Melting temperature (DSC, 10K/min | DIN EN 11357-1)

325 °C

**General properties**

Density (23°C | ISO 1183)

1,5 g/cm³

Humidity absorption (70°C, 62% r.H. | ISO 1110)

0,7 - 0,9 %

Molding shrinkage (flow | ISO 294-4)

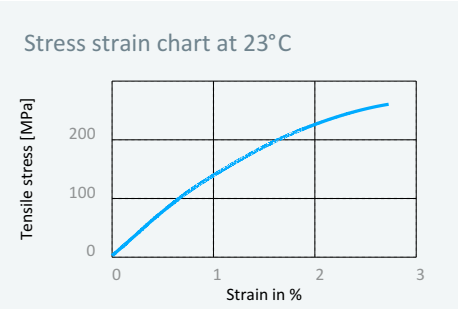
0,1-0,3 %

Molding shrinkage (transverse | ISO 294-4)

0,5-0,7 %

Disclaimer:

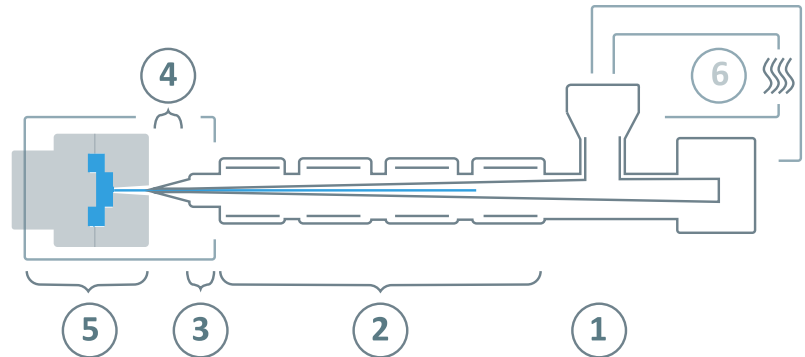
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AKROMID® T5 GF 40 8 natural (6463)**Processing information**

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only dehumidifying or vacuum dryers. Extensive drying can cause filling problems and surface defects.



⑥	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	120°C
	Processing moisture	0,02 - 0,1%
①	Feed section	60 - 90°C
②	Temperature zone 1 - Zone 4	320 - 350°C
③	Nozzle temperature	330 - 350°C
④	Melt temperature	330 - 350°C
⑤	Mold temperature	120 - 160°C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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